Descriptive Analysis of Solar Panel Data

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1 introduction

Since 2022, my parents have had solar panels installed on their roof. This document provides a descriptive analysis of the solar panel data and examines several practical questions related to electricity production, household consumption, grid usage, battery use, and seasonal variation.

A common assumption is that owning a solar panel system automatically leads to substantial financial benefits. However, the actual value of such a system depends on how much electricity is produced, how much of it is used directly, how much is stored in the battery, and how much still has to be imported from the public grid. This analysis therefore takes a data-based look at the performance of the system and evaluates how efficiently the generated solar energy is used.

The main goal is to better understand the cost-efficiency of the system. At the same time, this can also be interpreted as a measure of ecological efficiency: the more locally generated solar energy is used by the household, the less dependent the household is on externally supplied electricity.

A second motivation is to assess the current degree of grid dependence. In particular, the analysis investigates how much of the household's electricity demand can be covered by solar production and battery storage, and how much still depends on electricity imported from the public grid. This is relevant not only from a financial perspective, but also from a resilience perspective: it helps answer whether the household could maintain its current electricity use during a major disruption of the German energy infrastructure.

1.1 structure of the data

Each row represents one aggregated 3-hour measurement interval. The timestamp marks the beginning of the interval. Based on the hour of the timestamp, each row is assigned to a daytime category.

```r
dat$daytime <- NA

dat[dat$hour %in% c(21, 0, 3), ]$daytime <- "night"
dat[dat$hour == 6, ]$daytime <- "morning"
dat[dat$hour == 9, ]$daytime <- "late_morning"
dat[dat$hour == 12, ]$daytime <- "midday"
dat[dat$hour == 15, ]$daytime <- "afternoon"
dat[dat$hour == 18, ]$daytime <- "evening"
```

The most important variables are:

- `production` [Wh]: energy generated by the solar panels.
- `consumption` [Wh]: energy consumed by the household.
- `battery_charge` [Wh]: energy used to charge the 10 kWh battery.
- `battery_discharge` [Wh]: energy supplied by the battery to the household.
- `grid_feedin` [Wh]: energy exported to the public grid.
- `grid_consumption` [Wh]: energy imported from the public grid.
- `battery_state_of_charge` [%]: current battery charge level.
- `direct_consumption` [Wh]: solar energy consumed directly by the household.

1.2 data cleaning

The raw dataset does not contain the same number of observations for every year-month combination. This is expected in part, because months have different numbers of days. As we already know a complete day contains eight observations. With this information we can calculate that January should contain $31 \times 8 = 248$ observations, whereas February contains either $28 \times 8 = 224$ observations or, in a leap year, $29 \times 8 = 232$ observations.

The following table shows the number of observations per year and month before cleaning:

```r
table(dat$year, dat$month)
```

```
##
##          1   2   3   4   5   6   7   8   9  10  11  12
##   2023   0   0 165 240 248 240 248 248 240 248 240 248
##   2024 248 232 248 240 248 240 248 226 240 248 240 248
##   2025 248 224 248 240 248 240 248 248 240 248 240 248
##   2026 248 221 248 240 208   0   0   0   0   0   0   0
```

The table shows that some months are incomplete. March 2023 is incomplete because data collection started during that month, so it is removed from the analysis.

```r
dat <- dat[!(dat$year == 2023 & dat$month == 3), ]
```

Other months contain only a small number of missing 3-hour intervals. For these months, the missing values are approximated using the mean from the same month, day, and hour in other years. In total, 65 out of 9256 are approximated.

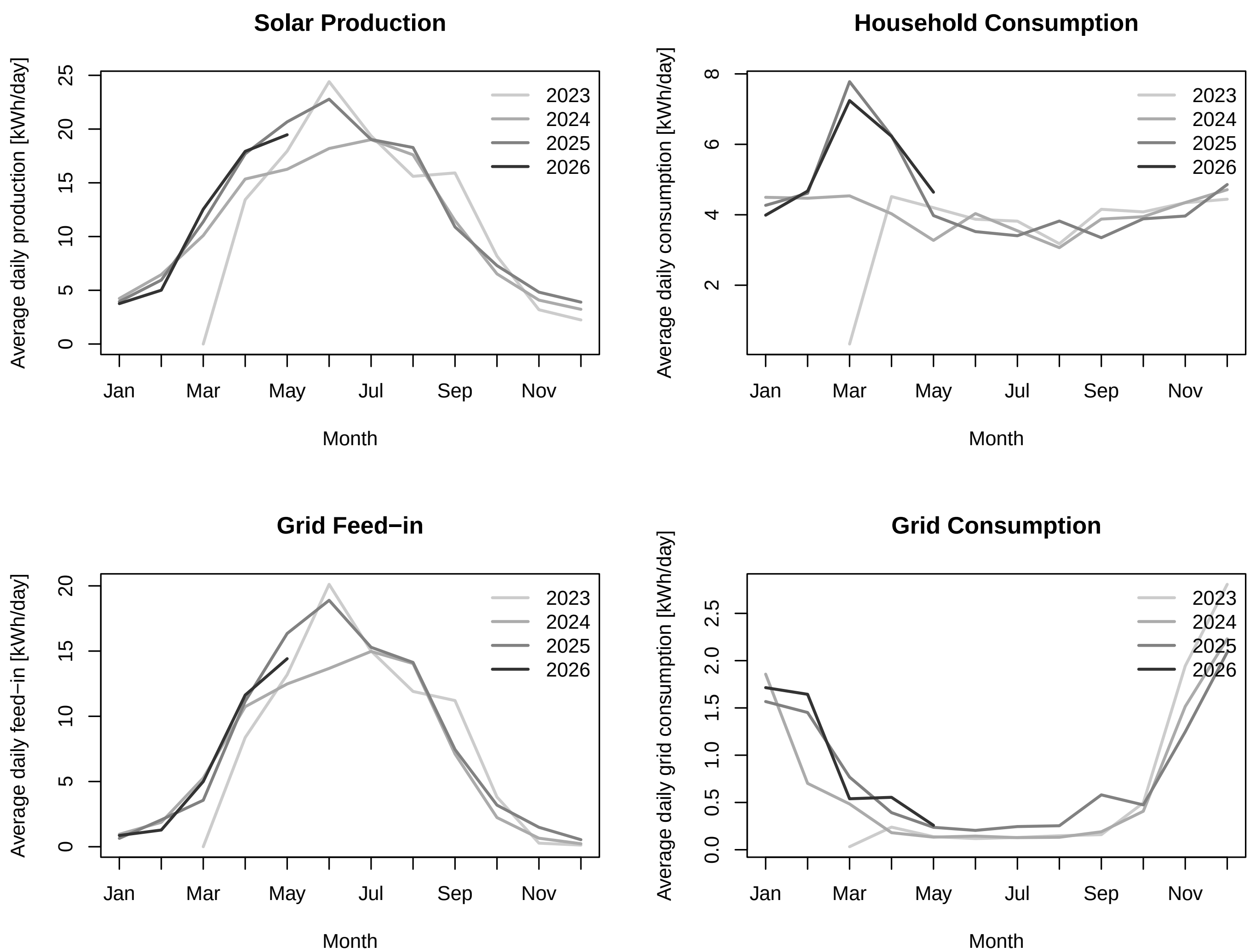
```r
table(dat$year, dat$month)
```

```
##
##          1   2   3   4   5   6   7   8   9  10  11  12
##   2023   0   0   0 240 248 240 248 248 240 248 240 248
##   2024 248 232 248 240 248 240 248 248 240 248 240 248
##   2025 248 224 248 240 248 240 248 248 240 248 240 248
##   2026 248 224 248 240 248   0   0   0   0   0   0   0
```

The resulting dataset preserves the correct calendar structure of the data while removing distortions caused by incomplete months or missing 3-hour intervals. This makes seasonal comparisons of absolute production and consumption values more meaningful.

2 descriptive analysis

First I want to look at the data from a yearly perspective, and decide if one can find categories like “seasons” in the data. Because months have different numbers of days, I use average daily values for monthly comparisons instead of simple monthly sums.



The plots show strong seasonality in the energy flows. Solar production and grid feed-in peak in the brighter months and decline sharply in winter. As a result, grid consumption remains very low during spring and summer, when solar availability is high, and rises again in autumn and winter as solar production declines.

Grid consumption and Solar Production / Grid Feed-in seem to be inverse. The Household Consumption stays on a consistent level all year round. One notable exception is the pronounced spike in household consumption in the 2025 and 2026 data. Although this pattern appears unusual at first glance, it has a specific explanation that will be addressed in the financial section of this paper.

2.1 defining solar seasons

Instead of using fixed calendar seasons such as the astronomical summer begin, I define seasons directly from the data as high-production and low-production solar months.

- **warm months**: the six months with the highest total solar production
- **cold months**: the six months with the lowest total solar production

Table 1: Estimated average monthly solar production and grid consumption, ranked by average monthly production

month	production	grid_consumption	n_years_observed	month_name	production_rank	grid_consumption_rank	g
6	653877	4671	3	June	1	12	v
7	593285	5200	3	July	2	11	v
5	576202	5954	4	May	3	9	v
8	532073	5513	3	August	4	10	v
4	483020	10257	4	April	5	7	v
9	382921	9303	3	September	6	8	v
3	351404	18617	3	March	7	5	c
10	227380	14135	3	October	8	6	c
2	164636	35566	3	February	9	4	c
1	123465	53761	3	January	10	2	c
11	121013	46908	3	November	11	3	c
12	96810	73243	3	December	12	1	c

Table 2: Total energy flows over the full observation period.

Energy flow	MWh
Solar production	13.98
Total consumption	5.03
Battery charge	2.57
Battery discharge	1.80
Grid feed-in	9.05
Grid consumption	0.87
Direct consumption	2.50

The table ranks months by total solar production. Months with ranks 1 to 6 are classified as warm months, while the remaining months are classified as cold months. To evaluate whether high-production months also tend to have lower grid consumption, I calculated the correlation between the production ranking and the grid consumption ranking, which is known as the Spearman Correlation. The resulting correlation is -0.97 and very high.

3 household energy balance

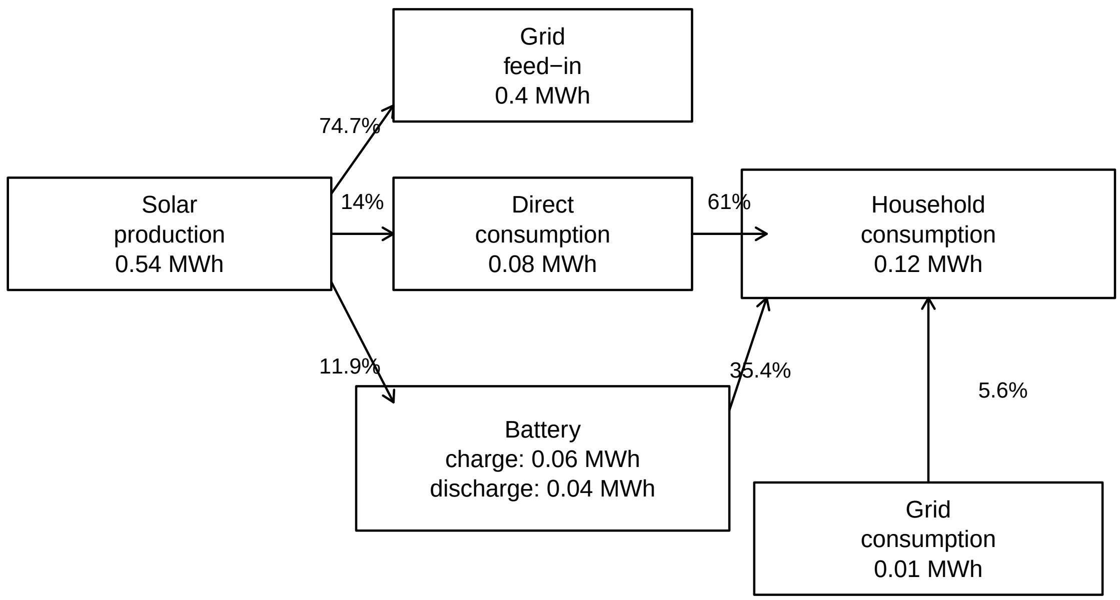
This section digs into the total energy flows over the full observation period. First we look at the the absolute energy amounts in MWh. Then we look at the energy flow, comparing the cold and the warm months.

3.1 total energy balance

The total values provide an overview of the household energy balance, but they hide important seasonal differences. Solar production is expected to be higher during the warm season, while grid consumption is usually more relevant during the cold season. Because of this, the energy flows were summarized separately for warm and cold months.

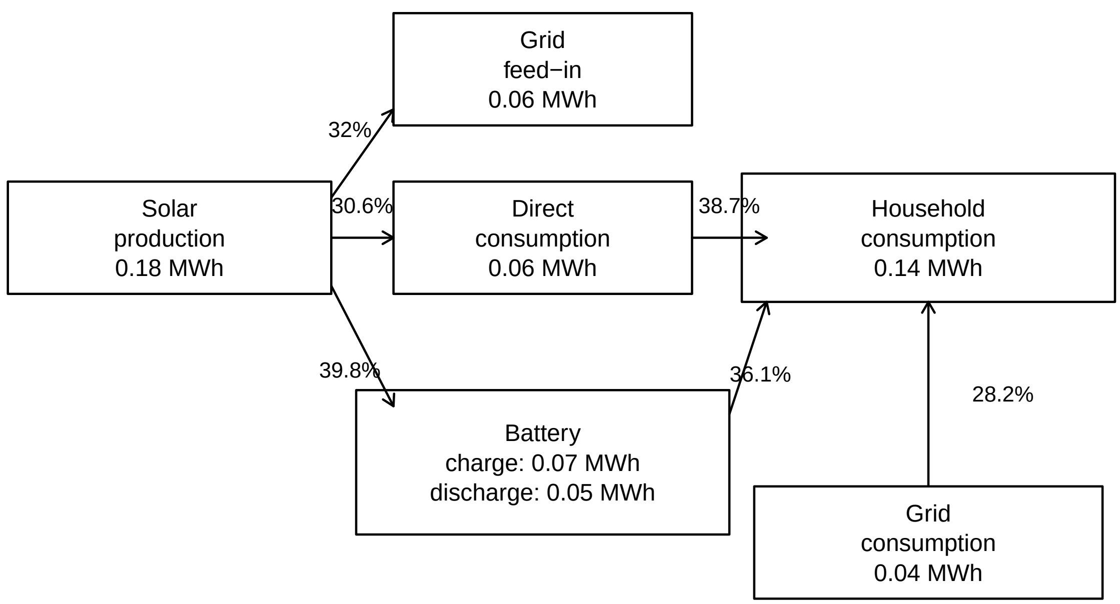
3.2 seasonal breakdown of household energy flows

Average monthly energy flows in high−production months



Left−side percentages are shares of solar production; right−side percentages are shares of household consumption.

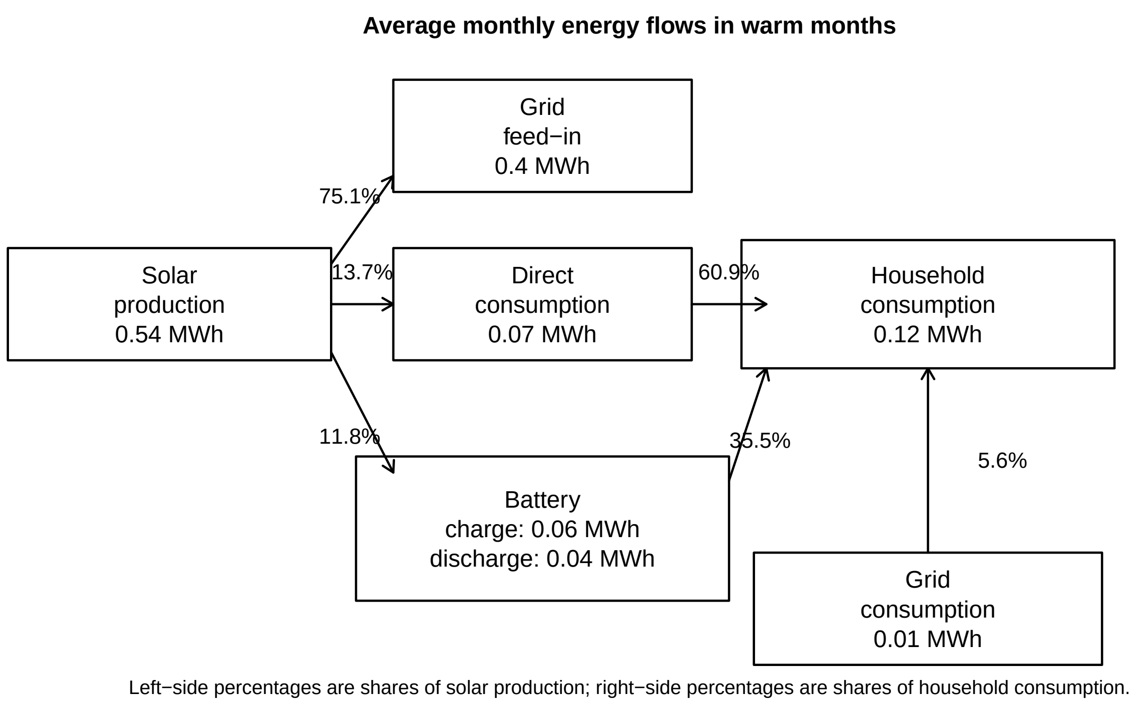
Average monthly energy flows in low−production months



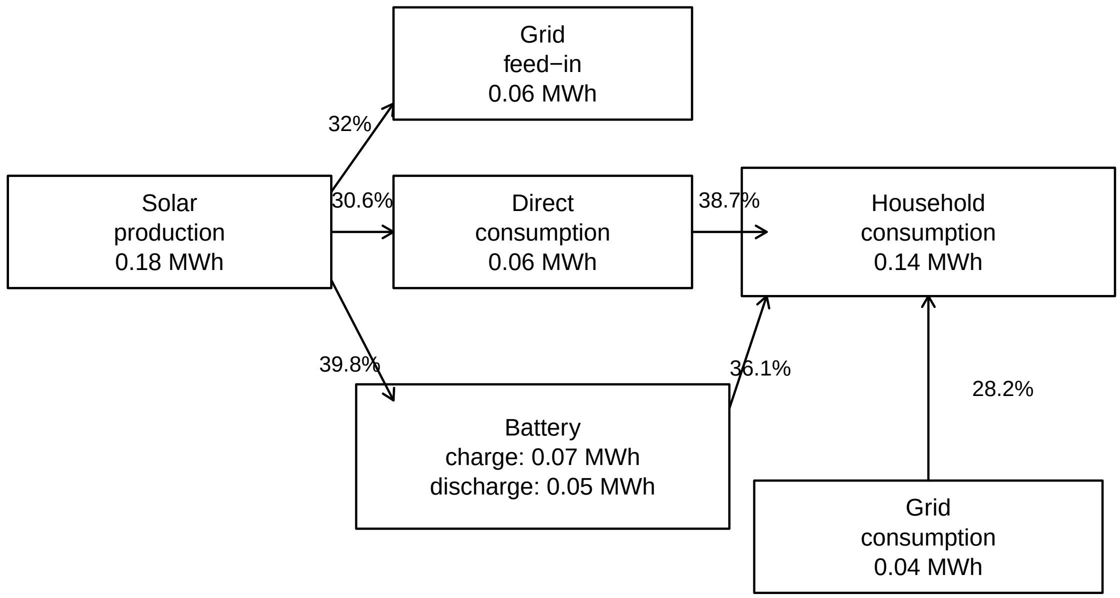
Left−side percentages are shares of solar production; right−side percentages are shares of household consumption.

```r
# flow_vars <- colnames(total)

flow_vars <- c(
  "production",
  "consumption",
  "battery_charge",
  "battery_discharge",
```

```
  "grid_feedin",
  "grid_consumption",
  "direct_consumption"
)

totals_monthly <- aggregate(
  dat[flow_vars], by = list(year = dat$year, month = dat$month),
  FUN = sum, na.rm = TRUE)

totals_monthly$season <- ifelse(
  totals_monthly$month %in% 4:9,
  "warm",
  "cold"
)

mean_by_month <- aggregate(
  totals_monthly[flow_vars],
  by = list(
    month = totals_monthly$month,
    season = totals_monthly$season
  ),
  FUN = mean,
  na.rm = TRUE
)

mean_by_season <- aggregate(
  mean_by_month[flow_vars],
  by = list(season = mean_by_month$season),
  FUN = mean,
  na.rm = TRUE
)
```



Average monthly energy flows in cold months



Left−side percentages are shares of solar production; right−side percentages are shares of household consumption.

3.3 financial implications of seasonality

As one might guess from the visualizations, there seem to be different money gains from the tow different seasons in the year.

lets see how much the system earned and spent every year for the cold and the warm season.

earning: 5 cent per kwh spending: 35 cent per kwh

ist es ausgeglichen

Here, the net balance is defined as total solar production minus total household consumption. A positive value means that the system produced more energy than the household consumed over the full observation period. what was the money balance?

The financial implications of grid exchange are strongly asymmetric because electricity bought from the grid is much more expensive than electricity sold to the grid. Assuming a feed-in compensation of 0.05 EUR/kWh and a grid electricity price of 0.35 EUR/kWh, one kilowatt-hour imported from the grid costs seven times as much as one kilowatt-hour exported to the grid earns.

In the warm months, grid feed-in generates about 395 EUR of revenue, while grid consumption costs only about 48 EUR. This results in a positive grid-related balance of about 347 EUR. In the cold months, however, grid feed-in generates only about 59 EUR, while grid consumption costs about 256 EUR, resulting in a negative balance of about 197 EUR.

This shows why direct consumption and battery use are financially more valuable than feeding electricity into the grid. Each kilowatt-hour used directly by the household avoids buying electricity for 0.35 EUR, whereas feeding the same kilowatt-hour into the grid would only earn 0.05 EUR. Therefore, using solar electricity locally is worth about 0.30 EUR more per kWh than exporting it.

total_warm$grid_feedin * 1000 * 0.05

total_cold$grid_feedin * 1000 * 0.05

total_warm$grid_consumption * 1000 * 0.35total_cold$grid_consumption * 1000 * 0.35

if feeding in and drawing were balanced total_warm$grid_feedin * 1000 * 0.35

```
net_energy_balance_mwh <- wh_to_mwh(
  sum(dat$production, na.rm = TRUE) - sum(dat$consumption, na.rm = TRUE)
)

net_grid_balance_mwh <- wh_to_mwh(
  sum(dat$grid_feedin, na.rm = TRUE) - sum(dat$grid_consumption, na.rm = TRUE)
)

data.frame(
  balance = c("Production minus consumption", "Grid export minus grid import"),
  energy_MWh = c(net_energy_balance_mwh, net_grid_balance_mwh)
)
```

```
##                         balance energy_MWh
## 1  Production minus consumption   8.943426
## 2 Grid export minus grid import   8.180250
```

Answer: The production-minus-consumption balance is about 8.9434258 MWh. The grid-export-minus-grid-import balance is about 8.1802497 MWh. This means the household produced substantially more solar energy than it consumed, but part of this surplus was exported because production and demand did not always occur at the same time.

3.4 7. What percentage of total solar production was used locally?

```
local_solar_use_wh <- sum(dat$production, na.rm = TRUE) - sum(dat$grid_feedin, na.rm = TRUE)
local_solar_share <- local_solar_use_wh / sum(dat$production, na.rm = TRUE)
local_solar_share_pct <- pct(local_solar_share)
local_solar_share_pct
```

```
## [1] 35.3
```